

Does the Galaxy Acceleration Scale Track Cosmic Dark Energy? A Non-Circular Cross-Scale $a_0(z)$ Test of de Sitter–Unruh Modified Inertia — the $z = 0$ Tie, the High- z BTFR Forecast, and Why M_{bar} Calibration (Not Rotator Count) Is Decisive

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Abstract

In a de Sitter–Unruh modified-inertia (MI) framework the low-acceleration scale a_0 is fixed to the cosmological horizon by $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} = 5.78881$, giving the pure-dark-energy-density (canonical) footing $a_0^{\text{canon}} = c^2\sqrt{\Lambda/32\pi} = 9.355 \times 10^{-11} \text{ ms}^{-2}$ or the total-density (ALT) footing $a_0^{\text{ALT}} = cH_0/Z = 1.1305 \times 10^{-10} \text{ ms}^{-2}$, two footings 20.9% apart that are carried on every number here. The framework does **not** rewrite Type Ia supernova standardization and gives **no** new distance formula — its cosmological background is General Relativity. Its distinctive content is that the dark-energy term in the Friedmann equation is not a free fit but the *galaxy* acceleration scale: $H^2(z) = \frac{8\pi G}{3}\rho_m(z) + Z^2 a_0^2(z)/c^2$, with $a_0(z)$ measurable from galaxy dynamics and Z a single posited constant. This ties a galaxy-kinematic quantity to a cosmic-distance quantity through $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\text{DE}}(z)}$ (equivalently $a_0 = c^2\sqrt{\Lambda_{\text{eff}}/32\pi}$ with $\Lambda_{\text{eff}} = 8\pi G\rho_{\text{DE}}/c^2$), i.e. $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$ — a bridge Λ CDM structurally lacks. It adds no dark-energy parameters beyond the GR background: the same $\rho_{\text{DE}}(z)$ that fits cosmic distances must fit the galaxy $a_0(z)$. The equality $a_0 \leftrightarrow \Lambda$ at fixed z is a definition and is circular; the *non-circular* test is the joint z -tracking of two independent datasets. We report a method + forecast **non-detection with a live cosmology-side hint**. (i) The $z = 0$ tie holds: Λ -blind SPARC $a_0(0) = 1.181 \times 10^{-10} (\pm 16\%)$ versus cosmic $(c/2)\sqrt{G\rho_{\text{DE},0}}$ gives ratio 1.26 (1.44σ , Planck H_0) or 1.17 (0.95σ , SH0ES H_0). (ii) The DESI DR2 $w_0 w_a$ posterior propagated through the framework relation yields $a_0(z=3)/a_0(0) \simeq 0.74$ (Pantheon+ 0.775, DESY5 0.737, Union3 0.707; a $\sim 2.2\sigma$ decline) — a consequence of the framework relation plus the DESI $w(z)$, not an independent detection. (iii) On the galaxy side the distinctive 0.60–0.75 decline is neither detected nor excluded: separability $S \simeq 0.8\sigma$. The Big Wheel — the cleanest available deep-MOND *candidate* at $z = 3.25$, though the v2 dynamical model finds it transitional ($y = g_{\text{bar}}/a_0 \approx 0.7\text{--}1.3$), not cleanly deep-MOND — gave, on the discovery-paper masses, $a_0(3.25)/a_0(0) = 1.31 (+0.93/-0.52)$, consistent with a constant, unable to separate the decline from flat, and (on those masses) disfavoring the ALT- $cH_0 \sim 5\times$ rise at $\sim 2\sigma$. **(v2)** A dedicated 2026 ALMA dynamical model (Quadri et al. 2026), *despite* sharper input

kinematics, does **not** tighten the single-object a_0 — it *empirically confirms* the central finding: a 0.37-dex stellar-mass ambiguity the authors flag as unresolved swings the inferred a_0 across ratio ~ 0.9 – 3.3 (0.4–4.7 across full systematics, $\sim 28\%$ of the high- M_* posterior unphysical). Under the authors’ fiducial dynamical mass the object itself sits at a $\sim 3\times$ rise (ratio 3.1 [1.9, 4.7]), so it no longer cleanly excludes the $\sim 5\times$ rise either — that $\sim 2\sigma$ rise-exclusion held only on the discovery masses. The single rotator cannot pin the decline; M_{bar} calibration, not the object, is decisive. The central finding is methodological: in the baryonic Tully–Fisher (BTFR) estimator a_0 is algebraically 1:1 degenerate with baryonic mass ($\Delta \log a_0 = -\Delta \log M_{\text{bar}}$ at fixed V), so the dominant M_{bar} systematic is common-mode and does **not** beat down with rotator count: with a $\gtrsim 0.08$ -dex correlated M_{bar} floor the test never reaches 3σ at any N . The decisive lever is M_{bar} calibration to $\lesssim 0.05$ dex common-mode, not the number of galaxies, and the a_0 -separable observable that breaks the degeneracy is the full rotation-curve / radial-acceleration transition, not the BTFR zero-point. The value of a_0 and the constant Z are posited, not derived; the a_0 magnitude inherits Z , so only the ratio/tracking is under test.

1. Introduction

1.1 The framework and the horizon scale

Modified inertia proposes that the inertial response of a body to a force weakens below a critical acceleration a_0 , rather than the gravitational field being modified. In the de Sitter–Unruh realization, an accelerated observer in a universe with a cosmological horizon sees a floor in the vacuum’s response set by the horizon temperature, and the transition scale is tied to the horizon:

$$a_0 = \frac{cH_\Lambda}{Z} = \frac{c^2\sqrt{\Lambda/3}}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.78881,$$

with $H_\Lambda \equiv c\sqrt{\Lambda/3}$ the de Sitter (pure dark-energy) Hubble rate. The Planck 2018 value $\Lambda = 1.089 \times 10^{-52} \text{ m}^{-2}$ gives the **canonical footing** $a_0^{\text{canon}} = c^2\sqrt{\Lambda/32\pi} = 9.355 \times 10^{-11} \text{ m s}^{-2}$ (pure ρ_{DE}). A second **ALT footing** replaces the pure- Λ density by the total present-day critical density, $a_0^{\text{ALT}} = cH_0/Z = 1.1305 \times 10^{-10} \text{ m s}^{-2}$. The two differ by 20.9%; both are carried on every dimensional number below. Nothing here derives the value of a_0 , the constant Z , or the sign of the inertial correction: those remain postulates. What the framework ties to the horizon is the *scale* of a_0 .

1.2 The parameter-free replacement of the free- w fit

The framework keeps General Relativity for the cosmological background and therefore has no native supernova formula: the light-curve $\rightarrow$ distance standardization is unchanged, and the Friedmann equation is the standard one. Its single

distinctive move is that the dark-energy term is not a free fit but the galaxy acceleration scale,

$$H^2(z) = \frac{8\pi G}{3} \rho_m(z) + \frac{Z^2 a_0^2(z)}{c^2}, \quad \frac{Z^2 a_0^2}{c^2} = H_\Lambda^2 = \frac{\Lambda c^2}{3} \quad (z=0),$$

so that, inverting $Z^2 a_0^2/c^2 = \frac{8\pi G}{3} \rho_{\text{DE}}$ with $Z^2 = 32\pi/3$, the dark-energy mass density is $\rho_{\text{DE}}(z) = 4a_0^2(z)/(Gc^2)$ and

$$\boxed{a_0(z) = \frac{c}{2} \sqrt{G \rho_{\text{DE}}(z)} = c^2 \sqrt{\Lambda_{\text{eff}}(z)/32\pi} \implies \frac{a_0(z)}{a_0(0)} = \sqrt{\frac{\rho_{\text{DE}}(z)}{\rho_{\text{DE},0}}}}$$

The SI form $\frac{c}{2} \sqrt{G \rho_{\text{DE}}}$ uses the dark-energy *mass* density ρ_{DE} (kg m⁻³); the equivalent geometric form uses $\Lambda_{\text{eff}} \equiv 8\pi G \rho_{\text{DE}}/c^2$ (units m⁻²). Both give $a_0^{\text{canon}} = 9.355 \times 10^{-11}$ at $z=0$ on Planck $\rho_{\text{DE},0}$; the two conventions must not be mixed ($c^2 \sqrt{\rho_{\text{DE}}/32\pi}$ with a *mass* density is dimensionally inconsistent). The redshift-tracking ratio $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$ is convention-independent.

On the supernova side this adds **no free dark-energy parameter** — against Λ CDM’s Ω_Λ (and, in the evolving-DE extension, w_0, w_a). The framework does not, however, produce a parameter-free dark-energy history from nothing: it *consumes* whatever $\rho_{\text{DE}}(z)$ the cosmic data prefer (itself the w_0, w_a fit to distances) and requires that the *same* function reproduce the galaxy-side $a_0(z)$. The content is therefore a cross-*dataset* constraint — one $\rho_{\text{DE}}(z)$ must simultaneously fit cosmic distances and galaxy kinematics — not a first-principles prediction of $w(z)$. At $z=0$ the framework is identical to Λ CDM by construction ($Z^2 a_0^2/c^2 = \Lambda c^2/3$). It differs from Λ CDM only if $a_0(z)$ evolves — that is, only if the galaxy acceleration scale changes with redshift in lock-step with the cosmic dark-energy density.

1.3 The non-circular cross-scale idea

Λ CDM gives no reason for a galaxy’s internal acceleration scale to track cosmic expansion; the framework forces exactly that link. Crucially, the equality of a galaxy a_0 and a cosmic Λ at a *single* epoch is a definition — one can always convert one into the other, and reading agreement there is circular. The genuine, non-circular test is whether **two independently measured functions of redshift** — a_0 from galaxy kinematics and ρ_{DE} from cosmic distances — track each other as z varies. That z -tracking is a prediction Λ CDM cannot make and the framework cannot avoid. The framework *inherits* the dark-energy history $w(z)$ (it does not predict $w(z)$); its content is the tie between the two datasets, so any decline it exhibits is a consequence of the framework relation

combined with the measured $w(z)$, not an independent statement about dark energy.

1.4 Credit for the kernel, and the status of this paper

The low-acceleration interpolation used to read a_0 from rotation curves, $\nu(y) = \sqrt{1 + 1/y}$ with $y = g_{\text{bar}}/a_0$, is Milgrom’s (1999, *Phys. Lett. A* **253**, 273, Eq. 9), in the original MOND context of Milgrom (1983). Milgrom’s coefficient was $2cH_\Lambda$ rather than cH_Λ/Z ; the deep-MOND limit $V^4 = a_0 GM_{\text{bar}}$ that anchors the galaxy-side measurement here is the baryonic Tully–Fisher relation (McGaugh 2005, 2012; Lelli, McGaugh & Schombert 2016). The distinctive content the present framework contributes is the cH_Λ/Z coefficient and the $a_0 \leftrightarrow \rho_{\text{DE}}(z)$ tie under test. This is a **method + forecast paper reporting a non-detection** with a live cosmology-side hint and a strategic finding; it is not, and does not claim to be, a detection. All load-bearing numbers are produced by committed, exit-0 scripts (Appendix A), and both footings are carried throughout.

2. The relation and why it is non-circular

Terminology. *Footing* — the choice of density that sets the a_0 *magnitude*: canonical (pure ρ_{DE} , 9.36×10^{-11}) or ALT (total present-day density, 1.13×10^{-10}); the two are 20.9% apart and cancel in any ratio. *Deep-MOND* — the low-acceleration regime $g \ll a_0$, where $\nu \rightarrow \sqrt{a_0/g}$ and $V^4 = a_0 GM_{\text{bar}}$ holds with fitted slope exactly 4. *BTFR zero-point* — the normalization of $V^4 = a_0 GM_{\text{bar}}$; in the deep-MOND limit that zero-point *is* a_0 , so reading $a_0(z)$ and reading the BTFR zero-point are the same operation. *Common-mode* — a systematic applied identically across a sample (one α_{CO} /IMF/dust prescription), which does not average down with sample size N .

2.1 The two-dataset bridge

Write the framework tie as a bridge between measurements made on wholly different physical scales:

$$\underbrace{a_0(z)}_{\text{galaxy kinematics: BTFR zero-point}} = \frac{c}{2} \sqrt{G \rho_{\text{DE}}(z)} = \underbrace{\frac{c}{2} \sqrt{G \rho_{\text{DE}}(z)}}_{\text{cosmic distances: SNe / DESI } \rho_{\text{DE}}(z)} .$$

The left side is read from the internal dynamics of individual galaxies (parsec–kiloparsec scales); the right side is read from the expansion history (gigaparsec scales). Λ CDM contains no relation forcing these to agree at all, let alone to co-evolve. The framework forces both their $z = 0$ equality and their joint z -dependence.

2.2 Definition versus test

The $z = 0$ equality $a_0(0) = (c/2)\sqrt{G\rho_{\text{DE},0}}$ is definitional: given a cosmic $\rho_{\text{DE},0}$ one *defines* the horizon a_0 , so a $z = 0$ match is a consistency knot, not an independent success. Its value is as the anchor the bridge extends from — if it failed badly the whole construction would be dead on arrival — but it carries no evidential weight beyond internal consistency. The non-circular content is entirely in the *slope*: does galaxy-measured $a_0(z)$ decline (or stay flat, or rise) in the same way the independently measured cosmic $\rho_{\text{DE}}(z)$ does? Because the two datasets never share inputs, a wrong background assumption shifts both together and cannot manufacture a spurious co-evolution (quantified in §4.4).

2.3 What the framework predicts, and what it inherits

At fixed z the framework prediction is fully specified once $\rho_{\text{DE}}(z)$ is chosen. Two reference cosmic tracks bracket the question:

track	(w_0, w_a)	$a_0(3)/a_0(0)$	$a_0(3.25)/a_0(0)$
flat- Λ ($w = -1$)	$(-1.00, 0.00)$	1.000 (flat)	1.000
DESI-DR2	$(-0.75, -0.90)$	0.712 (declining)	0.685
evolving (<i>ALT</i> $cH(z)$ <i>evolution model</i> , for contrast)	$a_0 \propto cH(z) =$ $cH_0 E(z)$	4.57 (rising)	4.99

with CPL history $\rho_{\text{DE}}(z)/\rho_{\text{DE},0} = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a z/(1+z)]$. The framework’s distinctive canonical-footing prediction is a **mild decline**, $a_0(3)/a_0(0) \approx 0.60$ – 0.75 , if the DESI-preferred evolving dark energy is real. If instead $w \rightarrow -1$ (flat), the framework reduces to Λ CDM at all z and the distinctive signal vanishes, leaving only the $z = 0$ tie. The third row is **not** the ALT $z = 0$ normalization ($a_0^{\text{ALT}} = 1.131 \times 10^{-10}$) but a distinct *evolution model* in which a_0 tracks the total-density Hubble rate $cH(z)$ and therefore *rises* steeply — the opposite sign — and it is the one branch the current galaxy data can already address (§4.3). The ALT $z = 0$ normalization itself, if it instead tracked $\rho_{\text{DE}}(z)$, would decline like the canonical footing and is *not* excluded; only the rising $cH(z)$ evolution is even approached by the current galaxy data — and, as §4.2 (v2) shows, only on the discovery-paper masses. The flat–DESI separation is only 0.147 dex at $z = 3$ (0.164 dex at $z = 3.25$); that small gap is what the galaxy data must resolve.

A theory-side caveat bounds the whole construction. The horizon relation $a_0 = cH_\Lambda/Z$ is derived for a *static* de Sitter horizon (constant Λ). Promoting it to $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\text{DE}}(z)}$ for an evolving equation of state ($w \neq -1$) is an **adiabatic / instantaneous-horizon ansatz** — the assumption that the inertia scale tracks the instantaneous dark-energy density — not a result derived from the de Sitter–Unruh construction, whose horizon is that of a constant- Λ background.

The z -tracking under test is therefore a test of this ansatz combined with the measured $w(z)$; a null would disfavour the ansatz, not the static-horizon relation it reduces to at $z = 0$.

3. The $z = 0$ tie and the cosmology side

3.1 The $z = 0$ tie

The Λ -blind galaxy $a_0(0)$ is taken from the gas-dominated SPARC BTFR zero-point measured without any cosmological- Λ input (the companion a_0 -line/ Λ paper, Zimmerman 2026, DOI 10.5281/zenodo.21419735): $a_0(0) = 1.181 \times 10^{-10} \text{ m s}^{-2}$ ($\pm 16\%$). The cosmic-side prediction is $a_0(0) = (c/2)\sqrt{G\rho_{\text{DE},0}}$ with $\rho_{\text{DE},0} = \Omega_{\text{DE}} 3H_0^2/(8\pi G)$:

footing	cosmic $a_0(0)$	$a_{0,\text{SPARC}}/a_{0,\text{cosmic}}$	tension
Planck ($\Omega_{\text{DE}} = 0.685$, $H_0 = 67.4$)	9.36×10^{-11}	1.26	1.44σ
SH0ES- H_0 ($\Omega_{\text{DE}} = 0.685$, $H_0 = 73.0$)	1.01×10^{-10}	1.17	0.95σ

The tie **holds at** $\sim 1\sigma$ on the canonical (Planck- H_0) footing and slightly better (0.95σ) at the local H_0 . This is by construction — the definitional anchor — and holds. It re-states that the framework enters the high- z test on solid $z = 0$ footing, and carries no evidential weight beyond that internal consistency.

3.2 The cosmology side: DESI DR2 propagated

On the cosmic side there is no free dark-energy parameter to fit — the framework consumes whatever $\rho_{\text{DE}}(z)$ the data prefer. We propagate the DESI DR2 (2025, arXiv:2503.14738) w_0w_a CDM posterior (BAO + CMB + SNe; representative marginals with w_0 - w_a correlation ≈ -0.87 ; Λ CDM excluded at 2.8σ /Pantheon+, 4.2σ /DESY5, 3.8σ /Union3) through $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$ (Appendix A, `desi_posterior_a0z.py`):

SNe combination	$a_0(3)/a_0(0)$	[16%, 84%]	$P(a_0 \text{ declines})$	decline signif.	in 0.60–0.75
DESI+CMB+Pantheon+	0.77	[0.683, 0.878]	97.9%	2.0σ	38%
DESI+CMB+DESY5	0.75	[0.651, 0.833]	99.3%	2.5σ	51%
DESI+CMB+Union3	0.70	[0.599, 0.831]	98.3%	2.1σ	48%

Across the three combinations the median is $a_0(3)/a_0(0) \simeq 0.74$ with a $\sim 2.2\sigma$ decline, and all three posteriors overlap the framework’s 0.60–0.75 band. This is a genuine, *live* cosmology-side hint: **if** the DESI evolving-dark-energy signal is real, the induced $a_0(z)$ decline falls squarely inside the framework’s 0.60–0.75 band. It is not, however, an independent test — it is a re-expression of the DESI $w(z)$ result through the framework relation, the *same* DESI information seen twice, and it therefore **cannot be added to the galaxy-side** χ^2 as if it were a second measurement. It dissolves entirely if the DESI signal regresses to $w = -1$. It sharpens the *target* for the galaxy side; it does not by itself confront it.

4. The galaxy-side $a_0(z)$ from the high- z BTFR

4.1 The estimator

On the galaxy side, $a_0(z)$ is read from the BTFR zero-point at each redshift. In the deep-MOND regime $V^4 = a_0 GM_{\text{bar}}$ is exact (the fitted slope is forced to 4), so at fixed circular velocity

$$\frac{a_0(z)}{a_0(0)} = \frac{(V_z/V_0)^4}{M_{\text{bar},z}/M_{\text{bar},0}}, \quad \text{equivalently} \quad \Delta \log a_0 = -\Delta \log M_{\text{bar}}|_{\text{fixed } V}.$$

Reading $a_0(z)$ is therefore the same operation as reading the BTFR zero-point. The estimator is valid **only** where the fitted slope is genuinely 4, i.e. for objects deep in the low-acceleration regime $g \ll a_0$.

4.2 The one clean high- z datum: the Big Wheel at $z = 3.25$

A single clean deep-MOND high- z rotator is currently available — the Big Wheel (Wang et al. 2025, *Nature Astronomy* **9**, 710; $z = 3.2514$; arXiv:2409.17956), a large, low-acceleration disk whose outer rotation curve satisfies the deep-MOND condition. Monte-Carlo over its velocity and baryonic-mass systematics ($V_{\text{rot}} = 280 \pm 31 \text{ km s}^{-1}$, $\sigma_{\text{int}} = 61 \text{ km s}^{-1}$ with asymmetric-drift correction; $M_{\star} = 1.7 \pm 0.7 \times 10^{11} M_{\odot}$, dynamically capped; $M_{\text{gas}} = 1.8 \pm 0.9 \times 10^{11} M_{\odot}$, $\times 1.36 \text{ He}$) gives $a_{0,\text{eff}} = 1.54 (+1.10/-0.61) \times 10^{-10} \text{ m s}^{-2}$, i.e.

footing	$a_0(3.25)/a_0(0)$
SPARC 1.181×10^{-10}	1.31 (+0.93/−0.52)
canonical $0.936 \times 10^{-10} (\rho_{\text{DE}}/cH_{\Lambda})$	1.65 (+1.17/−0.65)
ALT $1.131 \times 10^{-10} (\rho_{\text{tot}}/cH_0)$	1.37 (+0.97/−0.54)

On **every** footing the discovery-mass ratio is consistent with a constant (1.0) and with the DESI decline (0.68), while it **disfavors the rising $cH(z)$ evolution**

model ($\sim 5\times$) **at** $\sim 2\sigma$ (banked posterior probability $\sim 1\text{--}3\%$). This was a real, if narrow, result *on the discovery-paper masses* — but it is **not robust to the v2 dynamical model** (Update below): under Quadri et al.’s fiducial stellar mass the object sits at ratio 3.1 [1.9, 4.7], itself a $\sim 3\times$ rise only $\sim 1.2\sigma$ from the $\sim 5\times$ track, so the rise-exclusion too is M_{bar} -hostage. (The claim that reaching $5\times$ needs M_{bar} wrong by ~ 0.7 dex assumed the discovery masses; the 0.37-dex dynamical-vs-SED ambiguity alone already spans most of that.)

What it cannot do is separate the 0.70 decline from flat. The M_{bar} systematic on a single object (≈ 0.3 dex in M_{bar} , propagating to ≈ 0.23 dex in the combined a_0 error) is itself larger than the 0.164-dex flat-vs-decline signal at $z = 3.25$. Worse, that 0.226-dex error is a *floor*: it holds the asymmetric-drift coefficient fixed at $\alpha = 3.4$ rather than Monte-Carloing it, whereas varying α moves the ratio from 0.95 ($\alpha = 0$) to 1.27 ($\alpha = 3.4$) — a further $\sim 0.1\text{--}0.15$ dex — so the true error is somewhat larger and the datum even less constraining (the omitted term also pushes a_0 *up*, i.e. anti-framework, so it does not rescue a “win”). We use the anti-framework choices at every fork: the dynamically capped stellar mass, which pushes the ratio *up* (the Monte-Carlo median is 1.31; the capped-mass point value alone is 1.27; the full-SED stellar mass would instead place it at 0.86, right on the DESI decline), and a high asymmetric-drift correction — and the datum still lands consistent with both tracks.

One further systematic works in the framework’s favour and is *not* applied to the central value, so the quoted ratio is conservative. The estimator $a_{0,\text{eff}} \equiv V^4/(GM_{\text{bar}})$ equals a_0 *only* in the deep-MOND limit $g_{\text{bar}} \ll a_0$. On the framework’s own interpolation $g_{\text{obs}}^2 = g_{\text{bar}}^2 + g_{\text{bar}}a_0$ one has exactly $a_{0,\text{eff}} = g_{\text{obs}}^2/g_{\text{bar}} = g_{\text{bar}} + a_0$, so the estimator *overstates* a_0 by g_{bar} whenever the rotation-curve point is not deeply in the MOND regime. For the Big Wheel ($M_{\text{bar}} \approx 4.1 \times 10^{11} M_{\odot}$, outer point at $R \sim 30\text{--}45$ kpc) $g_{\text{bar}} \approx 0.3\text{--}0.5 a_0$, so $a_{0,\text{eff}}$ is biased high by $\sim 30\text{--}50\%$: subtracting g_{bar} moves the datum from $a_{0,\text{eff}} = 1.54 \times 10^{-10}$ (ratio 1.31) toward $a_0 \approx 1.1\text{--}1.2 \times 10^{-10}$ (ratio ≈ 1.0), i.e. *closer* to flat and to the decline, and *further* from the rise. The “clean deep-MOND” label is therefore optimistic for this single transitional object — the correct treatment uses the full ν -kernel at the measured g_{bar}/a_0 , not the deep-MOND asymptote — but the direction of the correction only reinforces the “consistent-with-both” reading, and the actual outer-RC radius (hence g_{bar}/a_0) should be pinned before this datum is used quantitatively (the rise-exclusion, and the radius, are both revisited in the Update below, which supersedes this estimate).

Update (v2): the 2026 ALMA dynamical model. A dedicated dynamical model of the Big Wheel from deep ALMA CO kinematics (Quadri et al. 2026, *A&A*, arXiv:2605.04144; W. Wang among the authors) now supersedes the discovery-paper inputs: a mildly *rising* rotation curve reaching $V_{\text{rot}} \approx 314 \text{ km s}^{-1}$ at the outermost **measured** radius ≈ 16.5 kpc (the 20 and 35 kpc values are angular-momentum *extrapolations*, not measured RC points), $M_{\text{gas}} = 10^{10.76} M_{\odot}$, and — critically — a stellar mass ambiguous by 0.37 dex: the dynamical-model fiducial $M_{\star} = 10^{11.00} M_{\odot}$ versus a higher SED/half-mass

value $M_\star = 10^{11.37} M_\odot$ that the authors flag as *in tension* with the dynamics. Applying the framework’s exact a_0 -line kernel $a_0 = (g_{\text{obs}}^2 - g_{\text{bar}}^2)/g_{\text{bar}}$ at the measured 16.5 kpc point (the ν -kernel correction advocated above), the inferred a_0 swings from ratio 0.86 (SED $M_\star$, $a_0 \simeq 1.0 \times 10^{-10}$) to 3.25 (dynamical $M_\star$, $a_0 \simeq 3.8 \times 10^{-10}$) at central values — Monte-Carlo medians 1.28 [0.41, 2.60] and 3.11 [1.89, 4.71] respectively, with $\sim 28\%$ of the SED- $M_\star$ posterior unphysical ($g_{\text{bar}} > g_{\text{obs}}$: baryons over-predicting the rotation). The object sits at $y = g_{\text{bar}}/a_0 \approx 0.7\text{--}1.3$ — transitional, *not* deep-MOND — so a_0 is hypersensitive to M_{bar} , and the 0.37-dex stellar-mass ambiguity *alone* spans a factor ~ 3.8 in a_0 . This does **not** tighten the datum. It *empirically confirms* the paper’s central finding (§5) on the very object one would reach for: with new, dedicated dynamical data the single Big Wheel a_0 remains M_{bar} -hostage and cannot separate the framework’s ~ 0.68 decline at $z = 3.25$ (§2.3) from a strong rise. It is an honest downgrade of the one in-hand galaxy-side value — the discovery-based Monte-Carlo had placed it at 1.31, which sits inside but no longer characterizes the now much wider range — and it supersedes the g_{bar} -subtraction estimate above (which assumed the higher discovery-paper $M_{\text{bar}} \approx 4.1 \times 10^{11} M_\odot$; with the lower dynamical M_{bar} the correction can move a_0 either way, dominated by the $M_\star$ choice). Backed by `bigwheel_update.py` (Appendix A).

4.3 Why intermediate- z is not a clean a_0 probe

The temptation is to add the many $z \sim 0.5\text{--}2.3$ rotating disks (e.g. KMOS^{3D}). They do not help, for a physical reason: these massive rotators sit at $g \gtrsim a_0$, **not** deep-MOND, so their fitted BTFR slopes are 3.0–3.85, not 4, and the zero-point is then degenerate with the slope, the pivot, and baryon/dark-matter-fraction evolution. The inferred “ a_0 evolution” *flips sign with the analysis choice*:

- Übler et al. (2017, KMOS^{3D}, fixed local slope 3.75) $\rightarrow$ zero-point $\sim 0.3\text{--}0.4$ dex below local $\rightarrow$ naive map reads a_0 **rising** $\sim 2\text{--}3\times$;
- Sharma et al. (2024, free slope 3.21) $\rightarrow$ zero-point above local at their pivot $\rightarrow$ **opposite** sign;
- Di Teodoro et al. (2016) and Tiley et al. (2019, $V/\sigma > 3$) $\rightarrow$ **null**, no significant zero-point evolution.

The scatter across analyses (a factor $\sim 2\text{--}3$, in *either* direction) dwarfs the 0.70-vs-1.0 signal. These points are not valid a_0 readouts; where we include the $z \approx 2.3$ point in the joint fit we place it at its fixed-slope value 1.86 (rising — the *wrong* sign for the framework decline), so it counts against, not for, the framework.

4.4 The $a_0 = M_{\text{bar}}$ degeneracy and the mild distance dependence

Because $\Delta \log a_0 = -\Delta \log M_{\text{bar}}$ at fixed V , a mis-estimate of the baryonic mass propagates 1:1 into a_0 : the two are algebraically degenerate for this estimator. Rising molecular-gas fractions ($\sim 10\%$ at $z = 0$ to $\sim 50\%$ at $z = 2$), the CO-

to- H_2 conversion α_{CO} , dust, and IMF evolution all move the BTFR zero-point and are indistinguishable from an a_0 shift with the BTFR / point-mass method. This is the crux limitation of §5. The mild distance–cosmology dependence is separate and quantified: $M_{\text{bar}} \propto D^2$, and the ratio $D(\text{DESI})/D(\text{flat})$ shifts M_{bar} by ≤ 0.015 dex at $z = 1$ (smaller at higher z) — roughly $10\times$ below the signal and $15\times$ below the Big-Wheel error. A wrong background moves all points together and cannot manufacture a decline, so the test is **not circular in the fatal sense**: testing whether a_0 tracks ρ_{DE} is not the same as assuming the DE evolution.

4.5 Does the galaxy data show the decline?

No — neither way. Assembling the honest galaxy points (Appendix A, `confront.py`):

set	$\chi^2(\text{flat})$	$\chi^2(\text{DESI-decl})$	$\Delta\chi^2$	lean
all 4 points	1.08	3.05	−1.97	flat, weak
deep-MOND candidate only ($z = 0 +$ Big Wheel, discovery masses; transitional per §4.2/v2)	0.27	1.55	−1.28	flat, weak

The clean-only row ($z = 0 +$ Big Wheel) is the sole formally meaningful statistic; the “all 4 points” row is **illustrative only**, because the intermediate- z points are declared invalid a_0 readouts (§4.3, slope $\neq 4$, sign-contested) and the $z = 1$ entry is itself the median of a hand-picked literature straddle — quantities that cannot legitimately enter a likelihood. Both rows agree in direction: the data mildly prefer flat, $|\Delta\chi^2| \approx 1.3\text{--}2.0$ — not significant, and what lean exists in the intermediate- z points is toward *rising*, the wrong sign for the framework decline. The distinctive 0.60–0.75 decline is not detected; the data are consistent with both flat and the decline. The one branch the data disfavor is the rising $cH(z)$ evolution model ($\sim 2\sigma$, via the Big Wheel *on the discovery-paper masses* — a result the v2 dynamical model itself weakens, since under the fiducial dynamical mass the Big Wheel sits at a $\sim 3\times$ rise; §4.2).

5. The key finding: M_{bar} calibration, not rotator count, is decisive

The decisive question for any future test is how the significance scales. The separability statistic — the maximum sigma at which the *current* errors could ever tell flat from the DESI decline — is

$$S = \sqrt{\sum_i \left[\frac{\log a_0^{\text{flat}}(z_i) - \log a_0^{\text{DESI}}(z_i)}{\sigma_i} \right]^2}.$$

The current data give $S = 0.73\sigma$ (Big Wheel alone), 0.73σ (clean pair $z = 0 + 3.25$), and 0.80σ (all four points): **underpowered**, consistent with both tracks. Every galaxy point sits $\leq 1.2\sigma$ from *both* the flat and the declining track. This is no longer only a forecast: the 2026 ALMA dynamical re-analysis of the Big Wheel (§4.2, Quadri et al. 2026) demonstrates it directly — a 0.37-dex stellar-mass ambiguity on a single *dedicated-data* object swings its a_0 by a factor ~ 3.8 , exactly the M_{bar} -hostage behaviour this section predicts, and precisely why more objects at fixed M_{bar} calibration would not help. So more data are needed — but the crucial point is *what kind*.

The naive forecast assumes independent per-object errors that beat down as $1/\sqrt{N}$. The target signal is $-\log_{10}(0.70) = 0.155$ dex, requiring $\sigma_{\text{mean}} = 0.052$ dex for 3σ , hence $N \approx 4$ rotators at SPARC-like precision (0.10 dex) or $N \approx 24$ at realistic high- z per-object precision (0.25 dex). **This is wrong**, because the dominant M_{bar} error — α_{CO} , IMF, and dust prescriptions applied *identically* to a sample — is largely common-mode, not independent, and a common-mode error does not average down with N . With $\sigma_{\text{mean}}^2 = \sigma_{\text{ind}}^2/N + \sigma_{\text{cm}}^2$ and a correlated M_{bar} floor σ_{cm} :

M_{bar} common-mode floor	N for 3σ
0.00 dex	24
0.05 dex	~ 377
≥ 0.08 dex	never reaches 3σ at any N

Because realistic high- z M_{bar} calibration floors are ~ 0.08 – 0.15 dex (α_{CO} alone is a 0.1–0.2 dex systematic), the test is decisive **only if the common-mode M_{bar} systematic can be driven below ~ 0.05 dex** — a calibration problem, not a counting problem. This is the paper’s core methodological contribution: **the decisive observational lever is a uniformly calibrated M_{bar} ladder to $\lesssim 0.05$ dex common-mode, not the number of rotators**. A uniformly calibrated M_{bar} scale is worth more than dozens of noisy objects. (Both footings enter this identically — the forecast is expressed in ratios and dex, so the 20.9% footing choice cancels; it re-enters only the Big-Wheel absolute of §4.2.)

6. Forecast and the a_0 -separable observable

To reach 3σ on the 0.60–0.75 decline versus flat therefore requires (i) a common-mode M_{bar} calibration $\lesssim 0.05$ dex and (ii) a sample of ~ 20 –40 clean deep-MOND $z \sim 2$ –3 rotators, Big-Wheel-like, with gas-traced outer rotation curves (JWST/ALMA) or individually resolved deep-MOND curves (ELT/HARMONI). For fixed samples the per-object precision required is ≤ 0.16 dex ($N = 10$), ≤ 0.23 dex ($N = 20$), ≤ 0.33 dex ($N = 40$) — all under the assumption that these errors are independent, which returns the argument to the common-mode floor above. Clean deep-MOND disks like the Big Wheel are rare (the intermediate- z KMOS^{3D} rotators are $g \gtrsim a_0$, §4.3), so assembling 20–40 of them is itself an observational stretch.

The deeper methodological escape is to stop using the BTFR zero-point at all. Because a_0 and M_{bar} are 1:1 degenerate in the BTFR, no amount of BTFR data isolates a_0 from baryonic-mass evolution. The **a_0 -separable** observable is the *shape* of the rotation curve — the radial-acceleration-relation transition radius / the full RAR — which depends on a_0 at fixed M_{bar} and therefore breaks the degeneracy that the zero-point cannot. A high- z RAR-transition measurement (the acceleration at which the curve departs from the Newtonian-baryonic expectation) reads a_0 directly, without hostage to the gas-mass normalization. This is the observable to target: not more BTFR zero-points, but resolved high- z rotation-curve shapes.

7. Discussion

The cross-scale link is the substance here. Λ CDM contains no relation forcing a galaxy’s internal acceleration scale to equal, or to co-evolve with, the cosmic dark-energy density; the framework forces both. That makes the joint z -tracking of galaxy kinematics and cosmic distances a genuine, falsifiable prediction that is not available to Λ CDM — and it is non-circular precisely because the two datasets never share inputs. The present status is asymmetric between the two scales. On the **cosmology side** the test is live: the DESI DR2 evolving-dark-energy posterior, propagated through the framework relation, produces an $a_0(3)/a_0(0) \simeq 0.74$ decline (at $\sim 2.2\sigma$) that falls inside the framework’s 0.60–0.75 band — but this is a consequence of the framework relation plus the DESI $w(z)$, not an independent measurement, and it stands or falls with the DESI signal. On the **galaxy side** the test is a hostage to M_{bar} : the distinctive decline is neither detected nor excluded ($S \simeq 0.8\sigma$), the intermediate- z BTFR is systematics-dominated and sign-contested, and the one high- z datum — on the discovery-paper masses — could only disfavor the rising ALT footing, a result the v2 dynamical model itself weakens (§4.2). The honest reading is that the framework’s forced bridge is currently underpowered — not passed, not failed — and that the binding constraint is M_{bar} calibration rather than sample size.

Two caveats bound the interpretation. First, the a_0 magnitude inherits the

posited Z ; only the ratio/tracking is under test, so a detection of the decline would test the *relation*, not the *value* of a_0 or Z . Second, the $z = 0$ tie is definitional and carries no independent weight; the evidential content is entirely in the slope. The framework is honest both ways: at every fork in the Big-Wheel analysis the anti-framework option is taken, and the datum still lands consistent with both tracks; the adverse rising intermediate- z point is retained in the fit; and the forecast is corrected downward by the common-mode floor.

8. Conclusion

We have set out a non-circular cross-scale test of de Sitter–Unruh modified inertia: the framework replaces Λ CDM’s free dark-energy fit with the galaxy acceleration scale, $H^2 = \frac{8\pi G}{3}\rho_m + Z^2 a_0^2(z)/c^2$, forcing $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$ — a bridge between galaxy kinematics and cosmic distances that Λ CDM structurally lacks. The $z = 0$ tie holds at $\sim 1\sigma$ (SPARC $a_0(0) = 1.181 \times 10^{-10}$ vs cosmic $(c/2)\sqrt{G\rho_{\text{DE},0}}$; ratio 1.26 Planck- H_0 , 1.17 SH0ES- H_0), the definitional anchor. The cosmology side is live: the DESI DR2 posterior propagated through the relation yields $a_0(3)/a_0(0) \simeq 0.74$ at $\sim 2.2\sigma$ — a consequence of the framework tie plus the DESI $w(z)$, not an independent detection, and it dissolves if $w \rightarrow -1$. The galaxy side is underpowered: the distinctive 0.60–0.75 decline is neither detected nor excluded ($S \simeq 0.8\sigma$); one $z = 3.25$ rotator, on the discovery-paper masses, only disfavored the ALT- cH_0 rise ($\sim 2\sigma$) — and the v2 ALMA dynamical re-analysis (§4.2) shows even that is M_{bar} -hostage, with the object at a $\sim 3\times$ rise under the fiducial dynamical mass. The central finding is methodological and redirects the observational strategy: in the BTFR estimator a_0 is 1:1 degenerate with M_{bar} , the dominant systematic is common-mode, and with a $\gtrsim 0.08$ -dex correlated M_{bar} floor the test never reaches 3σ at any rotator count. The decisive lever is M_{bar} calibration to $\lesssim 0.05$ dex common-mode, and the a_0 -separable observable is the full rotation-curve / RAR transition, not the BTFR zero-point. The value of a_0 and the constant Z are posited, not derived; both footings are carried throughout; and the test is reported honestly as a non-detection with a live cosmology-side hint and a strategic redirection of the future program.

Appendix A: The committed, exit-0 verification scripts

All numbers above are reproduced by three committed scripts (numpy/scipy only, exit 0, both footings), re-run 2026-07-18; the frozen public repository is left read-only and all outputs are written to `prep_2026/a0z_crossscale/`.

galaxy_a0z.py — the Λ -blind galaxy-side $a_0(z)$ from the high- z BTFR zero-point. Assembles the $z = 0$ SPARC anchor ($1.181 \times 10^{-10} \pm 16\%$), the sign-contested intermediate- z compilation (Übler+2017, Sharma+2024,

Di Teodoro+2016, Tiley+2019) with the deep-MOND-validity flag, and the $z = 3.25$ Big-Wheel Monte-Carlo ($a_{0,\text{eff}} = 1.54 (+1.10/-0.61) \times 10^{-10}$). Reports ratios so the footing cancels, and re-enters only the Big-Wheel absolute (SPARC/canonical/ALT = 1.31/1.65/1.37). Prints the forecast $N \approx 4$ (clean) / $N \approx 24$ (realistic) and writes `galaxy_a0z.png`.

bigwheel_update.py (v2) — re-extracts the Big Wheel a_0 from the Quadri et al. (2026) ALMA dynamical model via the a_0 -line kernel $a_0 = (g_{\text{obs}}^2 - g_{\text{bar}}^2)/g_{\text{bar}}$ at the measured 16.5 kpc point, for both stellar-mass scenarios ($M_\star = 10^{11.00}$ dynamical vs $10^{11.37}$ SED), with Monte-Carlo errors; demonstrates the M_{bar} -hostage swing (ratio $a_0/a_0^{\text{SPARC}} = 0.41\text{--}4.71$, $\sim 28\%$ unphysical at high $M_\star$), the empirical confirmation of §5.

confront.py — confronts the galaxy $a_0(z)$ against the flat- Λ and DESI-DR2 evolving cosmic tracks. Produces the joint χ^2 (all-4: 1.08 vs 3.05; clean: 0.27 vs 1.55), the separability ($S = 0.73/0.73/0.80\sigma$), the $z = 0$ tie (1.26/1.44 σ Planck, 1.17/0.95 σ SH0ES), the both-footings Big-Wheel absolute, and the forecast table — the fixed- N precision requirements **and** the common-mode M_{bar} floor of §5 ($\sigma_{\text{cm}} = 0 \rightarrow N \approx 24$; $0.05 \rightarrow N \approx 377$; $\geq 0.08 \rightarrow$ never reaches 3σ at any N). Bottom line: $S(\text{all}) = 0.80\sigma \Rightarrow$ underpowered, consistent with both.

desi_posterior_a0z.py — propagates the DESI DR2 w_0w_a CDM posterior (three SNe combinations, correlated $w_0\text{--}w_a$ via a seeded Cholesky draw) through $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$, giving $a_0(3)/a_0(0) = 0.775$ (Pantheon+, 2.0 σ), 0.737 (DESY5, 2.5 σ), 0.707 (Union3, 2.1 σ); combined median $\simeq 0.74$, decline significance $\sim 2.2\sigma$, all overlapping the framework 0.60–0.75 band. The ratio is footing-, H_0 -, and M_B -independent (canonical ρ_{DE} footing by construction; Z cancels).

Supporting reconstructions from an independent lane (`prep_2026/a0z_from_sne/`, `crossscale.py`) confirm that Type Ia supernovae **alone** cannot measure $a_0(z = 3)$: the dark-energy residual $E^2 - \Omega_m(1+z)^3$ is a difference of large numbers, and the model-independent Gaussian-process reconstruction (Seikel, Clarkson & Smith 2012) is sign-pinned only to $z \approx 0.40$. The robust SNe deliverable is $a_0(0)$ ($\simeq 9.4 \times 10^{-11}$ at $H_0 = 67.4$, $\simeq 1.01 \times 10^{-10}$ at $H_0 = 73.0$), agreeing with SPARC at $\sim 1\sigma$; the $z = 3$ decline numbers come from the DESI/model references, not a SNe reconstruction — which is exactly why the two-dataset (galaxy $\times$ cosmic-distance) construction, and not SNe alone, is the test.

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